Bright Squeezed Light from a Singly Resonant Frequency Doubler

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We present a new technique for the efficient and simple production of amplitude-squeezed light with high coherent excitation, using a frequency doubler, in which only the fundamental wave is resonantly enhanced. Our theory predicts up to nearly 90% squeezing in the second-harmonic wave under ideal conditions. In our first experiment we measured up to 30% squeezing [having taken into account the limited quantum efficiency (63%) of our detection system] in a stable 50 mW beam. Compared to doubly resonant frequency doubling our new approach has considerable technical advantages.

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Second-harmonic generation was one of the first processes in nonlinear optics which were considered for the production of squeezed light. First calculations [1] showed that in principle squeezed light could be generated in a single path of a laser beam through a nonlinear medium; however, it turned out that due to the low nonlinearity of the available media the squeezing would be very low for reasonable power levels. Therefore a more complicated model system was investigated in which the nonlinear crystal is used between mirrors which form a cavity for both fundamental and second-harmonic wave. It was shown [2,3] that such a doubly resonant system should be able to produce strongly squeezed light at moderate power levels. A first experiment [4] demonstrated the principle and resulted in 13% squeezing in the fundamental wave. In later experiments [5,6] 40% squeezing in the second-harmonic wave from a monolithic frequency doubler has been measured in a transient regime for a few milliseconds because the required double resonance could not be stabilized. Recently Kürz *et al.* [7] presented a monolithic frequency doubler which produced 52% squeezing in the fundamental wave and could for the first time be actively stabilized to the double resonance for about 10 s; longer time intervals were not possible due to technical problems which can hardly be avoided without better nonlinear material.

In this paper we show that a singly resonant frequency doubler, where only the fundamental wave is resonantly enhanced and the second-harmonic light can directly leave the system at one output mirror, cannot only be used for efficient frequency doubling, but can also produce strongly amplitude-squeezed light in the second-harmonic wave. A first obvious technical advantage of this system is that only one resonance has to be maintained; this allows for stable operation by using only one simple stabilization scheme. A second remarkable point is that the squeezed light can have a coherent excitation which is much stronger than in previous experiments. A power of 50 mW was achieved in our first experiment; higher powers could easily be obtained.

Before we present our theoretical data, we briefly outline the theory for this novel squeezing scheme. We use the same model as Ref. [8] for the description of the coupling to the nonresonant second-harmonic field, namely, a direct quadratic coupling to a bath. Thus from the point of view of the fundamental mode, the interaction is just two-photon damping, which is known not to produce high squeezing [3]. However, we now also have access to the output field associated with the nonlinear loss, and can use the general relations between internal and output statistics [9] to predict the squeezing properties of this second-harmonic output.

The equation of motion for the fundamental mode operator a can be written as

$$\dot{a} = -\mu a^\dagger a^2 + 2\sqrt{\mu}\, a^\dagger b_{\rm in} - \gamma a + \sqrt{2\gamma_0}\, a_{\rm in} + \sqrt{2\gamma_1}\, c_{\rm in}\,, \tag{1}$$

where μ is the two-photon damping rate and $b_{\rm in}$ the second-harmonic input field, γ_0 the input coupling for the fundamental mode and $a_{\rm in}$ the fundamental input field, γ_1 the intracavity loss rate at the fundamental (mainly absorption by the crystal) and $c_{\rm in}$ the quantum noise associated with this loss, and $\gamma = \gamma_0 + \gamma_1$ the total linear damping rate. We have assumed that the cavity is exactly resonant with the driving field. As boundary conditions we have

$$a_{\rm out} = \sqrt{2\gamma_0}\, a - a_{\rm in}\,, \tag{2}$$

$$b_{\rm out} = \sqrt{\mu}\, a^2 - b_{\rm in}\,. \tag{3}$$

With a vacuum input at the second harmonic, the semiclassical steady state is

$$\alpha = \frac{\sqrt{2\gamma_0}\,\alpha_{\rm in}}{\gamma + \mu|\alpha|^2}\,, \tag{4}$$

where Greek letters denote expectation values ($\alpha = \langle a\rangle$, $\beta_{\rm out} = \langle b_{\rm out}\rangle$, etc.). Without loss of generality, we may take $\alpha_{\rm in}$ and hence α to be real. The mean outputs are

$$\alpha_{\rm out} = \frac{\gamma_0 - \gamma_1 - \mu|\alpha|^2}{\gamma_0 + \gamma_1 + \mu|\alpha|^2}\,\alpha_{\rm in}\,, \tag{5}$$

$$\beta_{\rm out} = \frac{\sqrt{2\gamma_0\mu|\alpha|^2}}{\gamma + \mu|\alpha|^2}\,\alpha_{\rm in}\,, \tag{6}$$

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so that the maximum conversion efficiency of γ_0/γ is achieved when $\mu|\alpha|^2 = \gamma$. Relating the output power P_2 to the circulating fundamental wave power P_{1c}, we get

$$P_2 = \gamma_{\rm SHG}\, P_{1c}^2 = \frac{2\mu\tau^2}{\hbar\omega_1}\, P_{1c}^2 \tag{7}$$

with the round-trip time τ and fundamental wave photon energy $\hbar\omega_1$; this equation allows us to deduce the parameter μ from the conversion factor $\gamma_{\rm SHG}$, which depends both on the nonlinearity of the crystal material and the geometry of the resonator and is calculated for a monolithic standing wave resonator as used in the experiment in [10].

We now introduce fluctuation operators by

$$\delta a = a - \alpha \tag{8}$$

and similarly for the inputs and outputs. In terms of these, the equation of motion Eq. (1) can be linearized about the semiclassical operating point to give

$$\dot{\delta a} = -2\mu|\alpha|^2\delta a - \mu\alpha^2\delta a^\dagger + 2\sqrt{\mu}\,\alpha^*\delta b_{\rm in} + 2\sqrt{\mu}\,\delta a^\dagger\beta_{\rm in} - \gamma\delta a + \sqrt{2\gamma_0}\,\delta a_{\rm in} + \sqrt{2\gamma_1}\,c_{\rm in}\,. \tag{9}$$

The input field associated with the intracavity losses is always just noise, that is, $\langle c_{\rm in}\rangle = 0$. Equation (9) does allow for a coherent component to the second harmonic input, but in the rest of this paper it will be taken to be vacuum ($\beta_{\rm in} = 0$).

In frequency space, the solution for the fluctuations in the amplitude quadrature-phase operator,

$$X = a + a^\dagger\,, \tag{10}$$

is

$$\delta X(\omega) = \frac{\sqrt{2\gamma_0}\,\delta X_{\rm in} + 2\sqrt{\mu|\alpha|^2}\,Y_{\rm in} + \sqrt{2\gamma_1}\,Z_{\rm in}}{\gamma + 3\mu|\alpha|^2 - i\omega}\,, \tag{11}$$

where

$$Y_{\rm in} = b_{\rm in} + b_{\rm in}^\dagger\,,\quad Z_{\rm in} = c_{\rm in} + c_{\rm in}^\dagger\,. \tag{12}$$

The output amplitude fluctuations are then

$$\delta X_{\rm out}(\omega) = \frac{(\gamma_0 - \gamma_1 - 3\mu|\alpha|^2 + i\omega)\;\delta X_{\rm in} + 2\sqrt{2\gamma_0\mu|\alpha|^2}\,Y_{\rm in} + 2\sqrt{\gamma_0\gamma_1}\,Z_{\rm in}}{\gamma + 3\mu|\alpha|^2 - i\omega}\,, \tag{13}$$

$$\delta Y_{\rm out}(\omega) = \frac{(-\gamma + \mu|\alpha|^2 + i\omega)\;Y_{\rm in} - 2\sqrt{\mu|\alpha|^2}\,(\sqrt{2\gamma_0}\,\delta X_{\rm in} + \sqrt{2\gamma_1}\,Z_{\rm in})}{\gamma + 3\mu|\alpha|^2 - i\omega}\,, \tag{14}$$

at the fundamental and second harmonic, respectively. With vacuum or coherent inputs, the best squeezing obtainable in the fundamental output is

$$\langle\delta X_{\rm out}(0)\delta X_{\rm out}\rangle = 1 - \frac{\gamma_0}{3\gamma}\,, \tag{15}$$

when $3\mu|\alpha|^2 = \gamma$; this is the same result as for a two-photon absorber [3]. The result for the second harmonic is considerably more promising: we have

$$\langle\delta Y_{\rm out}(\omega)\delta Y_{\rm out}\rangle = \frac{(\gamma - \mu|\alpha|^2)^2 + \omega^2 + 4\mu|\alpha|^2 2\gamma}{(\gamma + 3\mu|\alpha|^2)^2 + \omega^2} = 1 - \frac{8\mu^2|\alpha|^4}{(\gamma + 3\mu|\alpha|^2)^2 + \omega^2}\,, \tag{16}$$

giving as a best case

$$\langle\delta Y_{\rm out}(0)\delta Y_{\rm out}\rangle = \tfrac{1}{9} \tag{17}$$

when $\mu|\alpha|^2 \gg \gamma$, i.e., when the doubler is operated at input powers higher than those where the conversion efficiency is maximized.

A qualitative explanation for the squeezing in the second-harmonic wave is the following: The second-harmonic fluctuations, mainly induced by the input vacuum fluctuations [$b_{\rm in}$ in (3)], influence the fundamental wave in such a way that the backaction from the latter tends to cancel the amplitude part of $b_{\rm out}$, but never by more than a factor of 9. This mechanism works approximately within the cavity bandwidth for the fundamental wave, however, with some power broadening: when the squeezing is good, it also has a large bandwidth.

Based on the considerations outlined above, we designed a monolithic standing-wave frequency doubler to reach the optimum conversion efficiency at low input powers. The resulting output powers were still low enough not to saturate the photodiodes of the homodyne detector which could not handle more than about 100 mW. The resonator (7.5 mm long) was made of MgO:$LiNbO_3$ with dielectric mirror coatings on the curved ($R = 10$ mm) end faces. The coupling mirror has a 99.6% reflectivity for the fundamental wave and a low reflectivity for the second harmonic, whereas the other mirror is highly reflective for both waves. Our design results in a strong nonlinearity and low round-trip losses (0.7%) for the fundamental wave, so that in the experiment the optimum conversion efficiency of 45% is reached with an input power of only about 50 mW. For the measured parameters of the device the theory predicts, e.g., 44% squeezing at 16 MHz in the second-harmonic wave for an input power of 100 mW.

The experimental setup is shown in Fig. 1. The pump laser is a diode-laser pumped monolithic Nd:YAG ring laser; it generates a clean TEM_{00} beam at 1064 nm. Its frequency can be tuned in a range of 1 GHz with a piezo acting on the laser crystal (fast tuning); furthermore, a tuning range of about 8 GHz is provided by the (slower) control of the temperature of the laser crystal. A Fara-

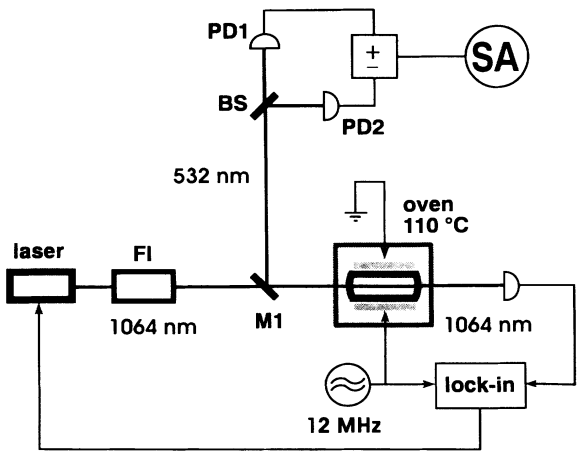

FIG. 1. Experimental setup. The pump beam at 1064 nm is mode matched to the monolithic resonator in the oven. A Faraday isolator (FI) suppresses backreflection to the laser. The lock-in feedback system locks the laser frequency to the fundamental wave resonance. The noise of the second-harmonic light, extracted from the pump beam with the dicroic mirror M1, is measured with a homodyn system [consisting of a 50% beam splitter (BS) and photodetectors (PD1 and PD2) and two spectrum analyzers (SA) HP 8591A].

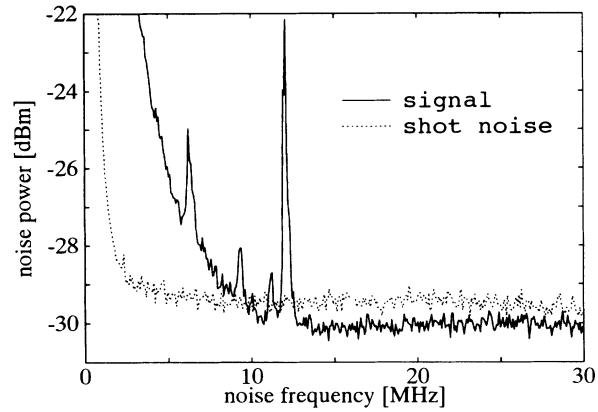

FIG. 2. Typical noise spectra, recorded at an input power of 80 mW with resolution bandwidth 300 kHz and video bandwidth 300 Hz. The squeezing can be seen at frequencies above and slightly below 12 MHz.

day isolator (FI) prevents the laser from being disturbed by pump light which is backreflected by the frequency doubler.

About 85% of the pump power is matched to the TEM_{00} mode of the frequency doubler crystal which is mounted in a temperature-stabilized oven operated near the phase-matching temperature of 110 °C. Two electrodes at the crystal allow for the application of a 12 MHz modulation with 5 mV_{pp} which is required for the active stabilization of the fundamental wave resonance. A lock-in technique is used to generate a servo signal for a feedback loop acting on the pump laser frequency by the use of both fast and slow tuning inputs of the laser.

The second-harmonic wave exits the crystal on the side of the pump beam and is separated from the latter by a dicroic mirror. It enters a balanced homodyne detection system without local oscillator which generates the photocurrent signals Φ_+ reproducing the intensity noise of the light beam and Φ_- as a reference for the corresponding shot-noise level [4–6]. The total efficiency of the detection system, depending on both the photodiodes (FND 100 from EG&G) and losses at several optical components (mirrors, a lens, and a polarizing beam splitter for the elimination of residual pump light), was measured to be (63±5)%. The detection electronics is designed for operation at frequencies between about 1 MHz and more than 30 MHz. The noise levels were recorded with two spectrum analyzers either in a 0 to 20 MHz span or in zero span at 16 MHz. Several tests confirmed that the homodyne system was balanced (more than 20 dB common mode suppression in the difference signal) and that the difference signal represents the shot-noise level.

Squeezing measurements were performed for pump powers between 10 and 100 mW. At each power level the oven temperature was carefully optimized for maximum nonlinear coupling. For pump powers below 50 mW optimum squeezing was obtained at the same time as the optimum conversion efficiency, but for higher powers these optimum temperatures did not coincide. This is as expected from the theory: If the pump power is larger than that for optimum conversion (with the maximum achievable nonlinearity), a reduction of the nonlinearity (by deviation from the phase-matching temperature) will increase the efficiency. A test with a 50% attenuator in the output beam confirmed the validity of the detection scheme: with the attenuator the squeezing was exactly half as much as without the attenuator.

Figure 2 shows typical noise spectra. The peak at 12 MHz results from the modulation which is required for the active stabilization. The peaks at lower frequencies are not yet fully understood; they seem to be related to small detunings. Above (and slightly below) the 12 MHz peak the squeezing shows up; for subsequent measurements we operated the spectrum analyzers in zero span mode at 16 MHz. In Fig. 3 we have shown the obtained squeezing (measured values corrected with the known detection efficiency) as a function of the pump power. The deviation between theoretical and experimentally obtained values is partly due to excess noise from the laser and, especially at higher powers, to thermal effects. The remarkable dip at about 50 mW pump power is clearly reproducible and seems to be related to a thermal instability of the system which also arises at these power levels. For higher pump power (about 75 mW) the system becomes bistable; in one of the operating points we observe low efficiency and strong excess noise, whereas in the other one there is squeezing (as plotted in Fig. 3), though probably less than would be obtainable without

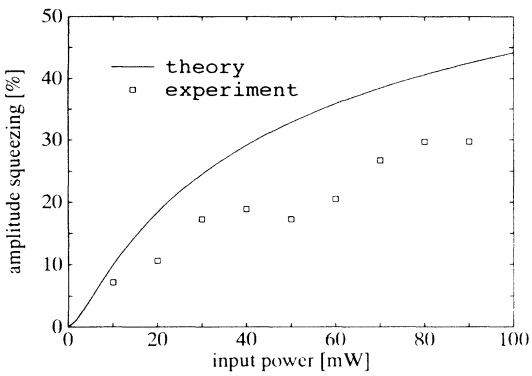

FIG. 3. Squeezing at 16 MHz as a function of pump power. The solid line shows the theoretical result based on measured parameters of the system, the squares the measured values (averaged over 1 s), corrected with the known detection efficiency.

the thermal effects. At pump powers above 90 mW the system is thermally unstable and cannot be operated at the optimum temperature.

A second singly resonant frequency doubler has been designed for optimum conversion efficiency around 100 mW pump power. The only difference from the other resonator is the lower reflectivity of the output coupler of 97.9%. Up to 82% conversion efficiency has been achieved [10]. The noise spectra are very similar to those shown in Fig. 2. At the same pump power, the noise suppression is smaller than for the other resonator, but the thermal problems are less severe so that the device can be pumped with up to 170 mW, where a second-harmonic beam with again 30% squeezing is produced, but this time with a power of 130 mW.

Considerable improvement of the results is to be expected from the use of $KNbO_3$, a material with higher nonlinearity. First calculations predict 70% squeezing at modest power levels. Therefore our new approach seems to make it possible to combine the advantages of a high degree and high bandwidth of squeezing, good stability, and high power level in one experiment. The singly resonant frequency doubler may be of considerable importance for applications of bright squeezed light, in particular highly sensitive spectroscopy [11] and the realization of low-noise optical amplifiers pumped with squeezed light [12], whereas the subthreshold optical parametric oscillator is the ideal source for squeezed vacuum [13], i.e., a squeezed beam without coherent excitation.

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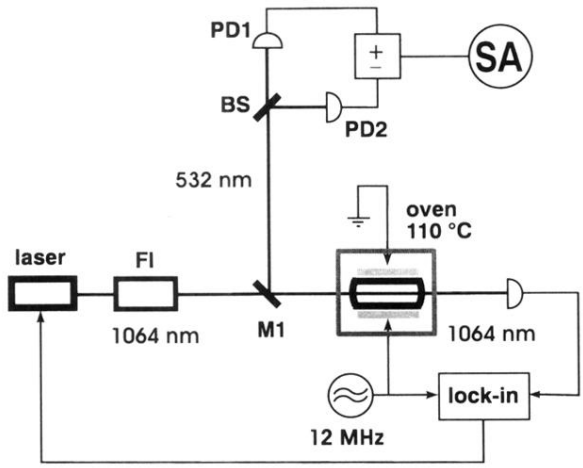

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